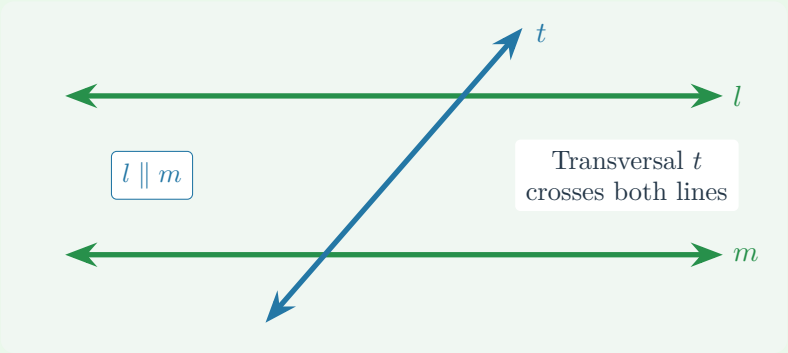
GANITA PRAKASH • CHAPTER 5

Class 7 Mathematics • Part 1

PARALLEL AND INTERSECTING LINES

Solve the textbook → see the rules → practise



Geometry is reasoning, not guessing

The notes cover **linear pairs, vertically opposite angles, perpendiculars, parallel lines, corresponding and alternate angles, and transversals**. NCERT questions are solved before the lesson explanations.

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NCERT Figure it Out: solved first

How to solve a diagram question

First mark which lines are **parallel** and which is the **transversal**. Then use one of these facts: a straight angle is 180°; vertically opposite angles are equal; corresponding and alternate angles are equal for parallel lines.

Figure it Out, page 108

P. 108 • All angle pairs in Fig. 5.3

Answer. With angles a, b, c, d around an intersection:

- **Linear pairs:** $(a, b), (b, c), (c, d), (d, a)$; each adds to 180°.
- **Vertically opposite pairs:** (a, c) and (b, d); each pair has equal angles.

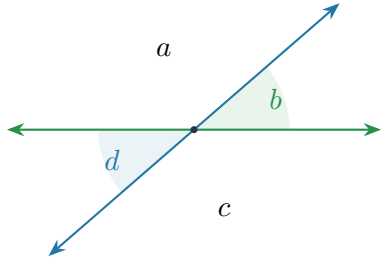


Figure it Out, pages 113–114

Q1 • Draw perpendiculars on dot paper (Fig. 5.10)

Answer. A correct answer is any line through a chosen grid point making a 90° **angle** with the given line. For a horizontal line draw a vertical; for a rising diagonal, construct a falling diagonal at 90°. **Rule used.** Place the right-angle corner of a set square at a dot and draw its other edge. Answers vary by which grid dots are chosen.

Q2 • Mark parallel and perpendicular pairs (Fig. 5.11)

Answer. Mark **parallel** segments with matching single/double arrows; use a little **square** where a horizontal meets a vertical at 90°. **(a)** Perpendicular pairs form a right angle. **(b)** Parallel pairs have the same direction and remain equally separated, however far extended.

Q3 • Make sets of parallel lines on dot paper

Answer. Many drawings are possible. For example, draw two horizontal segments between different rows of dots. Then draw another pair of diagonal segments, both rising by **one dot for every two dots to the right**. They have the same direction and are parallel.

Q4 • Make parallels to the segments in Fig. 5.12

Answer. For each segment a–h, copy its **direction or slope** to another part of the dot grid. Segments e, f, g, h can feel harder because they are slanted. One way: count the same *right steps* and *up/down steps* from the starting dot; join the new endpoints. **There is no unique drawing.**

Q5 • Which is parallel to a, b or c? (Fig. 5.13)

Answer. **Line c is parallel to line a**; b is not. **Rule used.** Line c has the same direction as a and the two do not meet if extended.

Figure it Out, page 119

Through A, construct a line parallel to l (Fig. 5.23)

Answer. Use a ruler and set square. **(1)** Align one set-square edge to l. **(2)** Hold a ruler against a second edge. **(3)** Slide the set square along the ruler until the first edge passes through A. **(4)** Draw along that edge. The new line is parallel to l. **Paper-fold alternative:** fold a line t through A perpendicular to l; then fold another line m through A perpendicular to t. Because l and m are both perpendicular to t, $l \parallel m$.

Figure it Out, pages 123–125

Q1 • Find the ten marked angles (Fig. 5.30)

Answer.

a	b	c	d	e
48°	52°	81°	99°	69°

f	g	h	i	j
48°	122°	75°	54°	97°

Rule used. Transfer angles using **corresponding or alternate-angle equality**; subtract adjacent angles from 180° when necessary. For the last combined-angle diagram, split the angle into the marked parts before comparing with parallel lines.

Q2 • Four angles called a (Fig. 5.31)

Answer. In the order drawn (top left, top right, bottom left, bottom right):

$$\boxed{138^\circ,\quad 118^\circ,\quad 105^\circ,\quad 23^\circ}.$$

Rule used. (i) Linear pair: $180 - 42 = 138$. (ii) Supplementary to 62°: 118. (iii) Use 110°, 35° and the adjacent straight/parallel-line angles: 105. (iv) The angle between the two oblique parallels and the vertical is $90 - 67 = 23$.

Q3 • Find x and y (Fig. 5.32)

Answer. **Left figure:** $x = 25^\circ$ and $y = 155^\circ$. As the vertical is perpendicular to the horizontal, $x = 90 - 65 = 25$. Angle y is supplementary to this acute angle: $180 - 25 = 155$.
Right figure: $x = 25^\circ$. The two slanted lines form angles of 53° and 78° with the *same horizontal direction*. The small angle between them is their difference: $\boxed{x = 78^\circ - 53^\circ = 25^\circ}$.

Q4 • Fig. 5.33, given $\angle ABC = 45^\circ$ and $\angle IKJ = 78^\circ$

Answer. $\boxed{\angle GEH = 45^\circ,\ \angle HEF = 57^\circ,\ \angle FED = 78^\circ}$.
Rule used. Transfer 45° and 78° using corresponding/vertically opposite angles; the three angles on the straight line at E add to 180°, leaving $180 - 45 - 78 = 57^\circ$.

Q5 • Fig. 5.34, three parallel lines, 55° at E

Answer. $\boxed{x = y = 125^\circ}$. **Rule used.** Each angle is supplementary to 55° at the corresponding intersections: $180 - 55 = 125^\circ$.

Q6 • Find $\angle NOP$ (Fig. 5.35)

Answer. $\boxed{108^\circ}$. **Rule used.** Draw a line through N and another through O parallel to LM and PQ. Use the 40° and 52° given angles and the 96° angle at N to find the remaining turn at O. The angle between ON and OP is $\boxed{96 + 52 - 40 = 108^\circ}$.

Understand the geometry

1. Intersecting and perpendicular lines

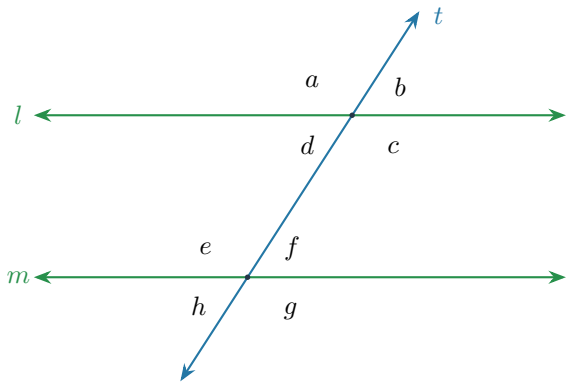
When two distinct straight lines meet at a point, they form **four angles**. Adjacent angles along a straight line form a **linear pair**, summing to 180°. Opposite angles across the crossing are **vertically opposite** and equal.

One angle tells you all four

If one angle is 120°, the opposite is 120° and both neighbours are 60°. If all four angles are equal, each is 90°: the lines are **perpendicular**.

2. Parallel lines and transversals

Parallel lines lie in the same plane and never meet, even when extended. A **transversal** crosses two or more lines.



Remember the angle families

If the two lines are parallel:

- **Corresponding angles** occupy matching corners and are equal (example $a = e$).
- **Alternate interior angles** are inside the parallels, on opposite sides of the transversal and are equal ($d = f$; $c = e$).
- **Co-interior angles** (same side, inside) add to 180° ($d + e = 180^\circ$; $c + f = 180^\circ$).

3. Proof matters more than measuring

A protractor may give 119° and 61° because of line thickness or measurement error. In ideal geometry the **linear-pair sum is exactly** 180°; it is proved using a straight angle, not guessed from appearance.

4. Other textbook reasoning checks

Quick worked examples and activities

- Fig. 5.2: if one crossing angle is $120°$, the other three are $60°, 120°, 60°$ around the point.
- Two distinct straight lines cannot meet twice.
- Fig. 5.6 (visual search): the pairs indicated as parallel include $c \parallel g$, $d \parallel f$, $e \parallel b$; the supplied answer page also groups a, i, h as same-direction lines.
- Eight distinct angle measures are impossible: vertical-opposite pairs are equal. Even **five** are impossible in Fig. 5.14, because $\angle 2 = \angle 4$.
- For parallel lines, if an alternate angle measures $120°$, the matching alternate angle is also $120°$.
- If $\angle 3 = 50°$ in Fig. 5.28, the adjacent angle is $130°$, so the corresponding $\angle 6 = 130°$.
- In Fig. 5.29: $\angle CAB = 55°$, $\angle ABC = 60°$, $\angle BCD = 120°$.

Paper-fold investigation (p. 111)

Answer. Opposite edges are **parallel**; adjacent edges are **perpendicular**. Repeated horizontal half-folds show **3, 5, 9** parallel lines (counting the two edges); in the ideal repeated-halving pattern, after n folds there are $2^n + 1$. A vertical fold meets them at $90°$. A new diagonal fold can be parallel to an earlier diagonal. In Fig. 5.8, corresponding folds $a \parallel p$, $b \parallel q$, $c \parallel r$.

Common mistakes and revision

Watch out

Wrong: All angles formed by a transversal are equal.
Correct: Only *specified pairs* are equal when the relevant lines are parallel. Adjacent supplementary angles usually differ.

Watch out

Wrong: Lines that do not meet *on the page* must be parallel.
Correct: Mentally extend them in both directions: if they would meet, they are not parallel.

One-page revision checklist

Relationship	Rule
Linear pair	$a + b = 180°$
Vertically opposite	$a = c$
Perpendicular lines	At their crossing: $90°$
Corresponding (parallels)	Equal
Alternate interior (parallels)	Equal
Same-side interior (parallels)	Sum $180°$

Self-test (with answers)

1. One angle at a crossing is $37°$. Its neighbour? $\boxed{143°}$.
2. Corresponding angles $76°$ and $76°$: parallel? *Yes, if the lines are cut by the same transversal in those positions.*
3. Same-side interior angle $112°$. Other? $\boxed{68°}$.
4. Can perpendicular lines also be parallel? *No; they meet at a right angle.*